

OFFLINE RESEARCH PROTOCOL

A Rigorous Experiment for Finding EV+ Sports Markets

How to compare EuroLeague, NCAAB, WNBA and tennis without confusing a profitable-looking backtest with a deployable edge.

Primary objective	Select and validate one league-market-book-time cell
Primary entry	T-60 minutes
Closing benchmark	T-5 minutes
Initial staking	Flat one-unit paper stakes
Research status	Research signals, not guaranteed profits

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1 Executive conclusion

The experiment should not begin by asking which league produced the highest backtest ROI. It should begin by asking which market has reliable point-in-time data, several independent prices, executable bookmakers and enough volume to test honestly.

The initial market tournament should compare four research cells:

Priority	League and market	Role in the experiment	Main risk
1	EuroLeague match winner, including overtime	Leading deployment candidate	May have insufficient independent Spanish-executable books
2	Predeclared high-coverage NCAAAB match winner subset	High-volume challenger	Conference heterogeneity and US-focused execution
3	WNBA match winner	Clean smaller-market challenger	Short season and limited sample
Control	ATP/WTB match winner	Sharp binary benchmark	Competitive prices and retirement settlement differences

The expected first choice is EuroLeague, but the data - not preference - must decide. Challenger tennis and top European domestic basketball remain future hypotheses because the present odds source does not provide the necessary public coverage.

The core idea can be summarized as:

EV+ is a property of league x market x bookmaker x decision time x jurisdiction

A league may be less visible than the NBA and still be difficult to beat. Lower visibility can coexist with wider margins, copied prices, tiny limits, missing results and rapid account restrictions.

2 What the experiment is trying to discover

The project is not primarily forecasting who will win. It is testing whether one bookmaker occasionally offers a price that is too generous relative to a robust, de-vigged market consensus available at the same decision time.

For a candidate price O and a conservative calibrated win probability p_{safe} :

$$\text{EV}_{\text{safe}} = O_{\text{executable}} \times p_{\text{safe}} - 1$$

A candidate is actionable only when all three elements are defensible:

- $O_{\text{executable}}$: the price was genuinely available to the user, not merely present in an API response.
- p_{safe} : a calibrated probability reduced for uncertainty, not the raw disagreement among bookmakers.
- Point-in-time validity: every input existed at or before the decision timestamp.

The experiment must therefore answer two different questions:

Question	Evidence required
Is the market researchable?	Coverage, independent prices, result matching, consistent settlement and sufficient sample size
Is the signal economically useful?	Calibration, positive closing value, positive untouched ROI, uncertainty bounds and real execution

3 Why the most obscure market is not automatically best

The attractive intuition is that fewer analysts and smaller trading teams should produce worse prices. That can be true, but pricing accuracy is only one part of deployable profitability.

Possible advantage	Corresponding operational cost
Slower bookmaker reaction	Price may disappear before an account can place it
Fewer quantitative models	Fewer independent bookmakers from which to build consensus
More outlier prices	More data errors, team-name mismatches and stale quotations
Less public attention	Lower limits and faster restrictions
Greater information asymmetry	The informed bettors present may represent a larger share of the market
High apparent backtest ROI	Low sample size and stronger selection bias

The desired market is not the smallest one. It is a mid-tier market with enough disagreement to create outliers and enough independent liquidity to make the consensus and execution credible.

Academic research on ten European domestic basketball leagues found no generic positive return at actually offered odds. Tennis research similarly shows that high predictive accuracy often fails to clear the margin. These findings do not eliminate individual mispricing; they eliminate the shortcut of declaring an entire smaller league soft.

4 The experiment at a glance

Phase	Purpose	Outcome
0. Preregister	Freeze the rules before seeing outcomes	Signed protocol and configuration
1. Coverage screen	Observe current market quality for six weeks	Pass or reject each market structurally
2. Select pilot	Choose using coverage and execution, not ROI	One historical dataset to acquire
3. Historical validation	Train, develop and test chronologically	One untouched holdout result
4. Prospective paper test	Run the frozen strategy forward	Real-time CLV and paper ROI
5. Small-capital execution	Measure obtainable prices, stakes and settlements	Operational validation
6. Calibrated sizing	Introduce constrained fractional Kelly	Only after probability and execution validation

Each phase is a gate. Failing a phase is a useful result: it prevents spending money on an untestable or non-executable market.

5 Phase 0 - Preregister before looking

Write a short protocol and freeze it before inspecting candidate profitability. At minimum, preregister the following:

- The four market cells and their exact settlement definitions.
- Primary decision time: T-60 minutes before scheduled start.
- Closing benchmark: T-5 minutes before scheduled start.
- Primary conservative EV threshold: 5%, unless changed before outcome inspection.
- Minimum quote freshness and bookmaker coverage.
- Primary de-vigging method and permitted development-stage challengers.
- Calibration family and minimum training sample.
- One-candidate-per-event selection rule.
- Exclusions for postponements, incomplete markets, stale quotes and ambiguous events.
- Primary metrics, confidence method and minimum final sample.

Any rule changed after the holdout is viewed becomes part of a new experiment. The old holdout can no longer be called untouched.

For NCAAAB, do not begin by labeling certain conferences as soft. Use the coverage phase to define a stable universe based only on data completeness, price independence and executable access. Freeze that universe before any outcome analysis.

6 Phase 1 - Six-week no-spend coverage screen

Collect current odds for every scheduled event in the four market cells. This phase measures research feasibility and should not rank markets using ROI.

For each event and snapshot, preserve:

- The raw provider response, unchanged.
- Provider event identifier, normalized participants and scheduled UTC start.
- Snapshot timestamp and each bookmaker's own update timestamp.
- Both outcomes and decimal prices for every complete bookmaker market.
- Overround, quote age, suspension state and missing outcomes.
- The price displayed on relevant Spanish operator sites for a manual sample.
- Final result, postponement, void and settlement status.
- Every exclusion and its explicit reason.

A market passes only if all structural gates are met:

Gate	Required level	Reason
Complete coverage	At least 5 books in at least 80% of events	Consensus needs breadth
Price independence	At least 3 effective price clusters	Brand count is not information count
Executable access	At least 2 usable candidate books	Reference prices alone cannot be monetized
Result matching	At least 98% deterministic matching	Settlement errors create false profit

Gate	Required level	Reason
Market semantics	Exactly 2 consistently settled outcomes	Avoid mixing overtime, regulation and draw markets
Price verification	API and operator price agree in manual checks	The stored price must be obtainable
Projected sample	Compatible with the frozen evidence standard	Small samples remain inconclusive

7 Phase 2 - Choose one market without outcome mining

Select the historical pilot using only market quality and deployability:

- 1 Complete event and bookmaker coverage.
- 2 Number of independent price clusters.
- 3 Reliability of event and result identity.
- 4 Quote freshness and provider latency.
- 5 Number of accessible candidate bookmakers in Spain.
- 6 Expected opportunities per month.
- 7 Likely stake capacity and settlement consistency.

Do not buy multiple datasets and then keep whichever produces the largest backtest. That transforms market selection into another model parameter and creates selection optimism.

If EuroLeague passes, it is the preferred first pilot because it produces the smallest engineering change from NBA. If it fails, use the frozen NCAAAB universe. WNBA is the clean engineering fallback but may be statistically inconclusive because of annual volume.

8 Phase 3 - Chronological historical validation

Partition the history into chronological blocks, preferably preserving complete seasons:

Block	Approximate share	Allowed use
Training	Earliest 60%	Fit calibration and estimate uncertainty
Development	Next 20%	Choose one specification and freeze it
Untouched holdout	Most recent 20%	Final evaluation exactly once

Random train/test splitting is prohibited because it mixes market regimes, bookmaker behavior and seasonal information across time.

Training data may be used to:

- Compare proportional and power de-vigging.
- Fit logistic or beta calibration.
- Estimate conservative probability lower bounds.
- Detect bookmaker families that move as one source.

- Set quote-age, overround and data-quality exclusions.

Development data selects one final specification. After that selection, every parameter, threshold and exclusion is frozen. The holdout is then evaluated once. A modification after seeing it requires a new future holdout.

9 The exact event-level decision sequence

Step	Operation	Safeguard
1	At T-60, select the latest record available at or before the decision time	No future snapshots
2	Validate freshness, completeness and market identity	Reject rather than silently repair
3	De-vig every bookmaker separately	Do not aggregate raw implied probabilities
4	Remove the evaluated bookmaker from its benchmark	Leave-one-bookmaker-out consensus
5	Take the median of valid peer probabilities	Reduce sensitivity to a single outlier
6	Apply the frozen chronological calibrator	Probability, not just ranking, must be reliable
7	Construct p_{safe} using the frozen uncertainty method	Dispersion is not automatically standard error
8	Verify the executable offered price	API availability is not execution
9	Calculate EV_{safe}	Use $O_{\text{executable}} \times p_{\text{safe}} - 1$
10	Apply the 5% threshold and retain one candidate per event	Avoid repeated-offer sample inflation
11	Record a flat one-unit paper stake	Keep sizing out of market selection

Calibration must be fitted on all eligible observations. Fitting only the high-EV selections creates selection bias and produces unstable probability corrections.

10 Closing value and outcome evaluation

At T-5, reconstruct a fresh de-vigged closing consensus using only information available by that timestamp. Define closing value as:

$$CLV = O_{\text{entry}} \times p_{\text{close}} - 1$$

Positive CLV means the entry price was better than the later market's fair assessment. Report mean CLV and the percentage of candidates with positive CLV.

CLV is a process diagnostic, not a substitute for realized return. The final untouched report should contain:

- Flat-stake return on investment and total turnover.
- Mean CLV and positive-CLV rate.
- Brier score, log loss and calibration by probability band.
- Candidate count and event count.
- Maximum drawdown and losing-run distribution.
- Results by bookmaker, season, odds band and signal strength.
- Performance after slippage, commission and quote-age stress tests.
- Performance after removing the most influential bookmaker.

Use an event- or time-block bootstrap rather than treating all bookmaker offers as independent observations. Repeated offers from the same game and neighboring events can share information and risk.

11 Frozen historical acceptance criteria

The untouched holdout passes only if every essential gate survives:

Dimension	Required evidence
Pricing process	Positive average de-vigged CLV
Economic result	Positive flat-stake ROI
Uncertainty	Predeclared lower 95% ROI confidence bound above zero
Calibration	Acceptable Brier score, log loss and reliability curve
Robustness	No dependence on one book, team, season or narrow odds band
Execution stress	Profit survives realistic slippage and missing-price assumptions
Sample	At least the frozen candidate requirement

The current project standard of 300 candidates must not be reduced after seeing the result. If a league cannot generate that sample, the correct label is inconclusive, not successful.

A profitable backtest with negative CLV, weak calibration or dependence on one period should be rejected even if headline ROI is attractive.

12 Phase 4 - Prospective paper trading

Run the frozen strategy forward in real time. Continue until 300 candidates settle or the relevant season ends. If the season ends first, retain the pilot label.

For every paper candidate, record:

- API price and timestamp.
- Price displayed by the operator at the attempted decision time.
- Whether the market was actually available.
- Maximum stake where observable.
- Any price movement between signal and verification.
- Closing consensus and CLV.
- Result, void or settlement discrepancy.
- Reason the candidate would have been rejected or unexecutable.

Do not repair the strategy during this phase. Log each issue for the next model version. Otherwise the prospective test becomes another adaptive backtest.

13 Phase 5 - Small-capital execution test

Only after the historical and prospective tests pass should the experiment use capital. Begin with fixed, very small stakes. This stage measures execution rather than model development.

Measure	Why it matters
Requested versus accepted stake	A theoretical edge with negligible capacity may have little economic value
Displayed versus executed odds	Small slippage can erase a thin edge
Rejected and changed-price bets	Signals may arrive after the opportunity is gone
Void and settlement rules	Different operators can grade the same event differently
Realized versus predicted EV	Tests whether calibration and execution survive contact with reality
Account restrictions	Determines the useful life and capacity of the strategy

Kelly staking is not part of this stage. Introduce capped fractional Kelly only after enough real execution data confirms probability calibration and price obtainability. Fractional Kelly cannot repair incorrect probabilities.

14 Decision matrix

Observed result	Decision
Coverage or settlement gates fail	Reject the market before purchasing history
Historical untouched test fails	Reject this specification
Historical passes but prospective CLV is negative	Reject; the historical edge probably did not persist
Positive CLV but ROI remains statistically inconclusive	Continue paper observation without staking
Statistical tests pass but prices are not executable	Reject operationally
Historical, prospective and execution stages pass	Begin calibrated small-scale deployment

Failure at any stage is not wasted work. It narrows the search space while protecting data budget and capital.

15 Required research outputs

Every experiment should be reproducible from raw inputs and a frozen configuration. Produce machine-readable and human-readable outputs:

- Raw snapshot manifest and provider provenance.
- Normalized games and results table.
- All valid offers with de-vigged probabilities.
- Candidates with p_market, p_calibrated, p_safe, EV and CLV stored separately.
- Exclusions table with an explicit reason for every rejected record.
- Flat-stake equity curve and drawdown series.
- Bookmaker, season, odds-band and signal-strength breakdowns.
- Calibration table and reliability plot data.
- Summary CSV and JSON containing configuration, sample and acceptance results.

Do not overwrite snapshots or omit failures. Provenance and exclusions are part of the experimental evidence.

16 One-page operational checklist

Stage	Checklist
Before collection	Freeze markets, timestamps, threshold, settlement, exclusions, metrics and sample
Coverage screen	Six weeks; 5 books in 80% of events; 3 price clusters; 2 executable books; 98% result matching
Pilot selection	Choose on coverage, independence, freshness, access, volume and capacity - not ROI
Historical data	Buy one bounded dataset only after a fresh cost and quota preflight
Training	Fit de-vigging, calibration and uncertainty on earlier data only
Development	Choose one final specification and freeze it
Holdout	Evaluate once; report CLV, ROI, calibration, drawdown, sample and robustness
Paper phase	No changes; verify displayed prices, availability, limits, settlement and CLV
Capital phase	Small flat stakes first; measure slippage and acceptance; Kelly later
Final label	Passed, failed or inconclusive - never retroactively loosen the gate

Immediate next step: begin the six-week T-60/T-5 coverage screen for EuroLeague, a broad NCAAAB observation universe, WNBA and ATP/WT. Do not optimize a profitability model or purchase history during this phase.

17 What transfers from the current code

The mathematical core is reusable, but the NBA application is not a universal sport switch. A new market must keep the generic calculations while adding a competition-specific boundary.

Reusable research logic	Required for every new competition
Point-in-time snapshot selection	Competition and season configuration
Two-outcome de-vigging	Team or player identity normalization
Leave-one-bookmaker-out consensus	Trustworthy schedule and result source
Median aggregation and conservative EV	Exact overtime, retirement, draw, void and postponement rules
Calibration and closing-value diagnostics	League-specific timing, calibration and minimum sample
Candidate, exclusion and performance reports	Provider coverage and executable-book verification

Changing only the provider sport key would silently retain NBA assumptions about teams, results, home and away identity, season structure and settlement. The safe implementation is a small adapter for each competition, with the generic two-way pricing core underneath.

18 Selected evidence and further reading

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8. Wilkens, S. Sports prediction and betting models in the machine learning age: The case of tennis. *Journal of Sports Analytics*. <https://doi.org/10.3233/JSA-200463>
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11. Practitioner caution: Tennis model with high accuracy and negative ROI. *Reddit, unaudited self-report*. [Link](#).
12. Practitioner caution: WNBA selected configurations fail in forward tracking. *Reddit, unaudited self-report*. [Link](#).

Research note

The literature and practitioner reports justify hypotheses and safeguards; they do not prove that any listed league is currently EV+. The experiment is designed to determine that prospectively, using the prices and operators actually available to the user.